

BEAUFORT WEST, South Africa

WMO: 687270

Lat: 32.347S

Lon: 22.573E

Elev: 2949

StdP: 13.20

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.2	38.6	8.9	9.7	67.7	12.3	11.5	65.7	35.2	64.9	31.5	65.5	5.6	50	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.6	97.2	62.1	94.2	61.6	91.7	61.0	67.4	81.8	66.2	81.2	65.1	80.5	12.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	98.4	71.2	61.9	92.4	71.0	60.4	87.4	70.5	33.6	81.7	32.6	81.1	31.7	80.0	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.9	23.8	20.4	DB	31.1	102.8	1.4	1.9	30.1	104.1	29.3	105.2	28.5	106.3	27.5	107.7	(4)
(5)			WB	28.4	69.9	2.2	3.2	26.8	72.2	25.5	74.0	24.3	75.8	22.7	78.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.0	75.5	75.7	71.6	65.3	59.9	55.2	54.6	56.2	60.2	64.9	68.9	73.2	(6)	
	DBStd	10.17	5.85	5.28	6.39	7.46	7.08	6.82	7.76	7.66	7.42	7.51	6.83	5.67	(7)	
	HDD50	95	0	0	0	1	6	23	38	21	5	1	0	0	(8)	
	HDD65	1552	4	1	17	86	191	304	334	292	184	95	38	6	(9)	
	CDD50	5585	790	719	670	460	314	178	182	214	310	464	566	718	(10)	
	CDD65	1568	328	300	223	95	35	9	12	21	39	93	154	259	(11)	
	CDH74	16586	3646	3130	2126	844	227	23	40	174	493	1165	1851	2867	(12)	
	CDH80	7201	1820	1483	872	254	33	0	1	31	153	451	787	1316	(13)	
(14) Wind	WSAvg	9.0	9.2	8.8	8.2	7.9	8.4	8.9	9.3	9.3	9.0	9.5	9.4	9.5	(14)	
(15) Precipitation	PrecAvg	9.4	1.0	1.3	1.7	0.9	0.6	0.4	0.4	0.6	0.4	0.8	0.9	0.9	(15)	
	PrecMax	16.7	4.4	5.5	5.4	3.9	3.8	1.0	2.0	3.0	2.0	2.9	2.8	4.2	(16)	
	PrecMin	4.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	(17)	
	PrecStd	3.9	1.2	1.4	1.6	1.0	0.8	0.3	0.6	0.8	0.5	0.8	0.7	1.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	101.2	99.6	96.2	91.1	83.9	76.9	78.0	83.7	90.0	95.1	96.8	99.4	(19)	
		MCWB	63.4	63.0	61.9	60.3	55.6	52.1	51.8	53.6	56.2	59.1	60.3	61.5	(20)	
	2%	DB	96.9	95.8	92.2	86.7	79.8	73.6	74.4	78.6	84.7	89.9	92.3	94.5	(21)	
		MCWB	62.7	62.7	61.5	59.2	54.1	50.6	50.6	51.4	54.0	57.8	59.0	61.3	(22)	
	5%	DB	93.5	92.3	88.7	82.9	76.4	70.9	71.4	74.8	79.7	85.6	88.3	91.3	(23)	
		MCWB	62.0	62.3	61.0	58.1	53.2	49.8	49.7	50.1	52.6	56.6	58.0	60.7	(24)	
	10%	DB	90.0	88.9	85.2	79.1	73.2	67.5	68.2	70.8	75.3	81.0	84.5	87.6	(25)	
		MCWB	61.3	61.8	60.1	56.9	52.5	48.8	48.7	48.6	51.1	55.2	57.5	59.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.5	69.7	68.1	64.5	59.4	54.6	54.3	56.9	59.5	63.1	64.7	67.2	(27)	
		MCDB	83.3	84.4	82.1	78.1	72.0	69.8	70.8	70.3	79.8	78.7	81.0	82.7	(28)	
	2%	WB	67.0	67.7	66.2	62.4	57.0	52.8	52.1	54.2	56.6	61.2	62.5	65.4	(29)	
		MCDB	83.2	80.9	79.7	77.2	71.3	67.8	69.9	70.9	74.3	79.0	81.5	82.2	(30)	
	5%	WB	65.6	66.5	64.7	60.9	55.7	51.4	50.7	52.1	54.7	59.4	61.1	63.8	(31)	
		MCDB	82.2	80.5	78.9	76.6	70.6	66.3	67.8	68.8	73.5	77.4	79.4	80.8	(32)	
	10%	WB	64.4	65.3	63.4	59.5	54.3	50.0	49.4	50.4	52.8	57.6	59.6	62.6	(33)	
		MCDB	81.1	79.8	77.4	74.0	69.2	65.0	65.6	67.7	71.9	76.2	78.2	79.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	25.6	24.3	22.1	19.7	18.1	18.0	18.4	20.0	22.3	23.8	25.1	25.7	(35)	
		MCDBR	30.9	29.6	27.2	23.9	22.2	20.5	20.5	24.2	28.7	30.5	31.4	31.1	(36)	
	5% WB	MCWBR	7.2	7.3	7.2	7.6	8.6	8.8	8.8	9.9	11.0	9.3	8.2	7.6	(37)	
		MCWBR	25.7	22.5	21.9	20.4	19.3	18.4	19.0	21.3	25.1	25.5	26.7	25.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.321	0.326	0.315	0.290	0.273	0.262	0.263	0.289	0.306	0.323	0.308	0.316	(40)	
		taud	2.562	2.546	2.569	2.641	2.672	2.671	2.653	2.538	2.466	2.458	2.559	2.564	(41)	
	Ebn at Noon		323	315	308	300	291	287	292	295	305	312	325	325	(42)	
		Edh at Noon	34	33	30	25	22	21	22	28	33	36	34	34	(43)	
(44) All-Sky Solar Radiation	RadAvg		2592	2325	1931	1465	1125	977	1072	1359	1805	2203	2535	2711	(44)	
	RadStd		93	95	91	63	48	47	53	65	94	87	106	140	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (6 neighbors)		N/A	N/A	+3.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page